What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Targeted Inhibitors Used in Cancer Therapy

Last updated May 2, 2025 · 12 min read · ScholarRx

Adapted by: Diptiman Bose, Trager Hintze

Editors: Jordan Finn, Russell Johnson

Adapted from: Targeted Inhibitors of Cancer Cell Function

Listen to this Brick
0:00 / 23:08 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Compare and contrast the key features between Tyrosine kinase inhibitors and monoclonal antibodies
2. Explain the mechanism of action of targeted inhibitors used in cancer therapy.
3. Explain the protein degradation
4. Describe the adverse effects associated with the use of targeted inhibitors and strategies for mitigating these effects.

CASE CONNECTION

You are a third-year clerkship student on rotation with LS's oncologist. LS, a 62-year-old female, was diagnosed with diffuse large B-cell lymphoma in January 2024. She was administered a combination of agents, including prednisone, doxorubicin, vincristine, rituximab, and cyclophosphamide. This was followed by a single dose of Rituximab until 2026. During a routine blood test, LS's AST and ALT levels were elevated, and she was referred to an oncologist. Upon examining the results, you notice that while her HBSAg (Hepatitis B surface antigen) antibody was negative before starting chemotherapy, it has now become positive. LS has a family history of chronic hepatitis B. LS is now concerned about why she developed Hepatitis B, and with her cancer history, she is very anxious.

How do you explain LS's current diagnosis? Consider your answer as you read, and we will revisit LS at the end of the brick.

GO TO CONCLUSION ↓

The development of targeted cancer therapies hinges on identifying molecular changes that drive the malignant progression of human cancers. An increasing number of drugs are being designed to inhibit oncogenic pathways responsible for uncontrolled cancer cell growth and survival. These targeted therapies can be combined with cytotoxic cancer drugs, such as alkylating agents and antimetabolites, to enhance their effectiveness.

In BOOM, we discussed various signaling pathways such as MAP kinase, JAK-STAT that are among myriad of signaling pathways regulating cell growth and proliferation. The primary pathways of cancer growth are oncogenic signaling pathways (Kinase pathways), reactions of the tumor microenvironment (angiogenesis), and evading the host immune response or surveillance.

The development of **small molecules** and **monoclonal antibodies** as tools targeting specific proteins or pathways have been key in management of several cancers. Small molecules can enter the cell and target intracellular signaling pathways (e.g., kinase pathways), whereas monoclonal antibodies can target plasma membrane receptors. The table below lists the differences between small molecule and monoclonal antibodies chemotherapy agents.

Table

SMALL MOLECULES	MONOCLONAL ANTIBODES
Targets are mostly intracellular	Targets are on cell surface
Have multiple targets	Generally specific for single antigen
Different selectivity	More selective
Broad spectrum of effects	Specific effects
Oral administration with CYP metabolism	Parenteral administration
Half lives 12-24 h: more frequent administration	Half lives days to weeks: less frequent administration

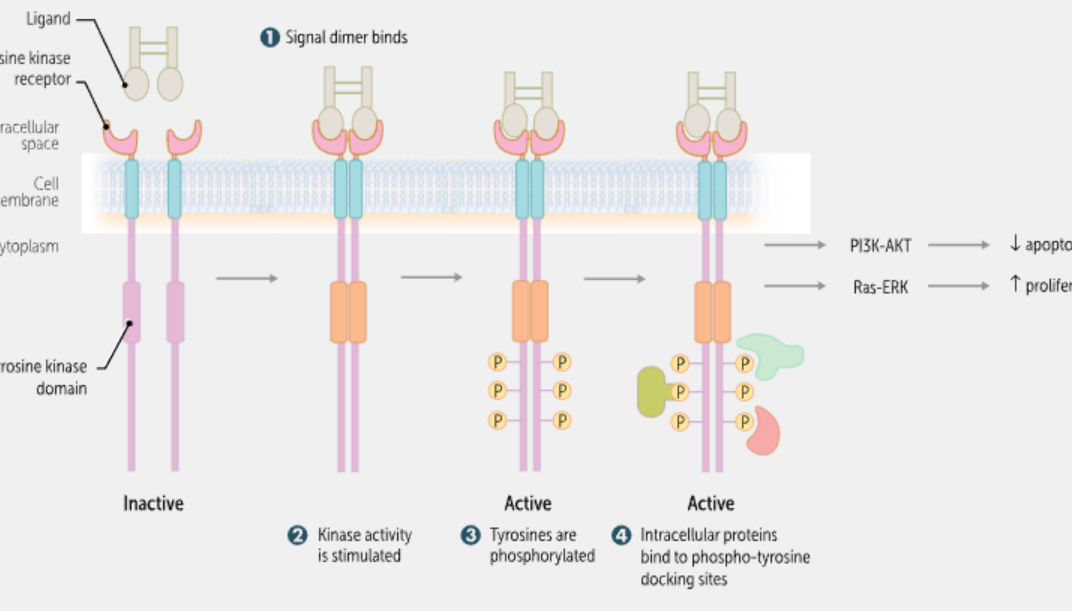
What Are Targeted Inhibitors of Cancer Cell Function? (LO#1)

The targets for small molecules and monoclonal antibodies are typically signaling cascades that are involved in cell proliferation. Recall from BOOM that signaling pathways activated by receptor tyrosine kinase are involved in such cellular function. In fact, a targeted approach to cancer therapy now commonly includes tyrosine kinase inhibitors (TKIs) and monoclonal antibodies raised against specific cellular proteins important in cancer cell propagation.

How Do Tyrosine Kinase Pathways Accelerate Cancer Growth?

Tyrosine kinase receptors are cell-surface receptors that initiate intracellular signaling pathways. These pathways are overactive in most cancers, leading to increased cell proliferation and decreased apoptosis (programmed cell death) characteristic of cancer cells.

Examples of this type of receptor include the epidermal growth factor receptor (EGFR), VEGF, and hepatocyte growth factor receptor. Structurally, they are transmembrane proteins with an extracellular portion and an intracellular domain. This type of receptor usually consists of two parts that connect when the ligand binds its site, forming a dimer. This union induces kinase activity. Kinases then phosphorylate the receptor, leading to its activation (Figure 1). Once activated, the phosphorylated receptor initiates two main pathways: the PI3K-AKT pathway and the Ras-ERK pathway. These pathways eventually lead to decreased apoptosis and increased cell proliferation, respectively.



QUIZ

Figure 1

Among these cancer cell targets and some drugs that are designed to block them are:

- Tyrosine kinase cell signaling receptors: crizotinib, erlotinib, imatinib, panitumumab, and trastuzumab
- Intermediate steps on cell signaling pathways: vemurafenib
- Proteasomes: bortezomib

How Do Tyrosine Kinase Inhibitors and Monoclonal Antibodies Dif

Because upregulated receptor tyrosine kinase (RTKs) pathways are crucial to cancer cell growth, survival, and angiogenesis, let's see how a drug might block them. Two agents, monoclonal antibodies and tyrosine kinase inhibitor drugs, achieve this blockage by two different mechanisms (Figure 2).

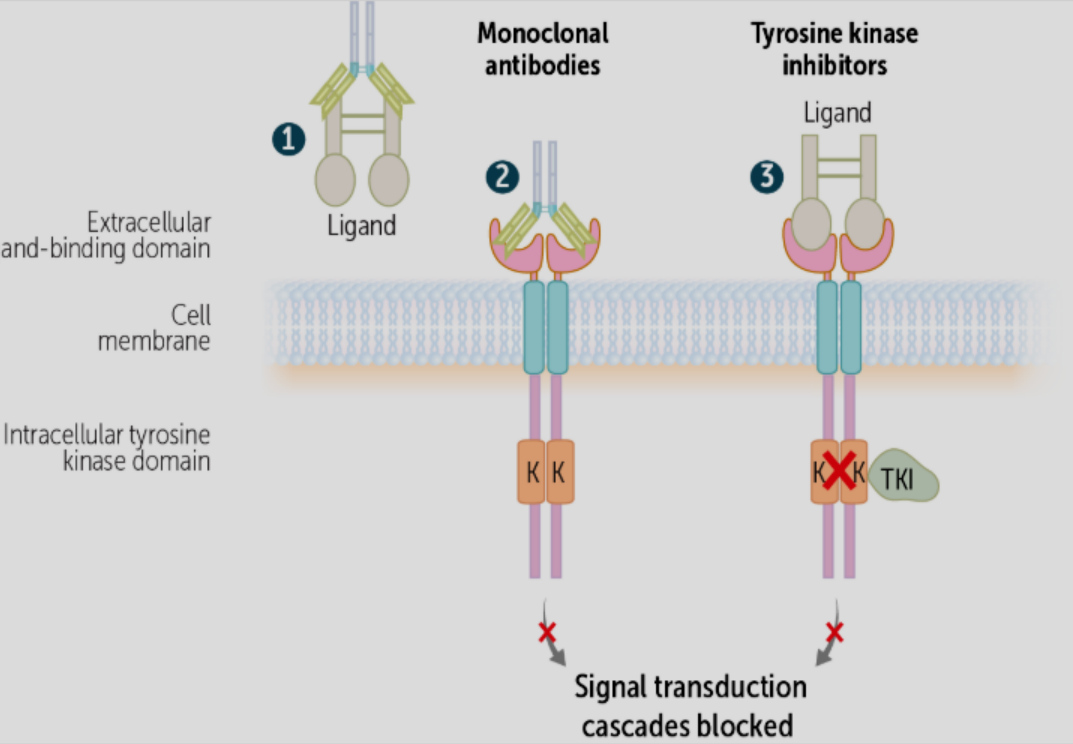

Monoclonal antibodies
Tyrosine kinase inhibitors
1
2
3
Ligand
Ligand
Extracellular ligand-binding domain
Cell membrane
Intracellular tyrosine kinase domain
K K
K K
TKI
Signal transduction cascades blocked

Figure 2

Mnemonic

MOnOclOnal AntibOdies attach on the Outside domain of the receptor.
TyrosIne kInase InhIbItors attach on the Inside domain of the receptor.

Tyrosine Kinase Inhibitors

Tyrosine kinase inhibitors (TKIs) are not antibodies, they are small molecules that chemically block receptor function, typically at its intracellular domain. Here, these molecules competitively inhibit adenosine triphosphate (ATP) binding, preventing the phosphorylation of the receptor and interrupting the signaling pathway. TKIs are not nearly as specific in their targeting, so they may interact with tyrosine kinase receptors other than the ones they were originally intended to target and can increase adverse effects. An easy method to recall TKIs is that they all end in ***-tinib***.

What Are Drugs that Target Tyrosine Kinase Signaling? (LO#2 and 4)

ImaTINIB

The Philadelphia chr between chromosome 9 and 22. The short Philadelphia chromosome (Ph1) is a fusion between the tyrosine kinase ABL (Abelson murine leukemia viral oncogene homolog) and Breakpoint cluster region (BCR). The new fusion gene, BCR-ABL is a constitutively active protein kinase, resulting in continuous and uncontrollable cell division.

CLINICAL CORRELATION

BCR-ABL drives the malignant phenotype of chronic myelocytic leukemia (CML)

Imatinib, the first TKI approved by the US Food and Drug Administration targets the *BCR-ABL gene*. Imatinib binds to the kinase site of the BCR-ABL gene competitively inhibiting the binding of ATP. Thus, the phosphorylation of downstream signaling mediated by the activation of RTK is inhibited.

What is the mechanism of action of Imatinib?

What are the potential adverse effects of Imatinib? **Fluid retention** is a common side effect. This can cause leg swelling and shortness of breath. Elevation of **liver function test enzymes, GI distress**, and muscle cramps can also occur. **Rash** developed from imatnib might require intervention as it could develop into SJS (Stevens-Johnsons syndrome).

DasaTINIB, PonaTINIB

Dasatinib is an TKI that inhibits the BCR-ABL gene. Unlike imatinib, it binds to both the active and inactive conformations of the Abl kinase domain and can overcome imatinib resistance resulting from mutations in the BCR-ABL kinase. It can prolong QT intervals and cause pleural effusion.

What is a common adverse reaction that can develop from Imatinib?

How do Monoclonal Antibodies work? (LO#2 and 4)

Monoclonal antibodies are developed in the lab to attach to a specific cell protein, often a cell-surface receptor. As you can see in Figure 2, monoclonal antibodies can target a cell surface receptor or the ligand of the cell surface receptor: either way, cell signaling is blocked. These antibodies are highly specific to a particular receptor or ligand, so crossover reactions (to a different, nontargeted receptor or ligand) are uncommon. After attachment, these antibodies can attract other substances to the cancer cell to assist in its destruction, such as additional chemotherapeutic agents. This can also stimulate inherent host cell immunity, such as antibody-dependent cellular cytotoxicity.

CLINICAL CORRELATION

Monoclonal antibodies are used for other purposes besides cancer therapy; examples include treatment of rheumatoid diseases and some antiviral protocols. You can recognize monoclonal antibodies by name because, although the prefix is variable, **the suffix is typically -mab**.

The syllables of these agents can differ, but they can reveal information about the agent, and their source. The information can be used to predict some of the adverse drug reactions and kinetics for the drugs.

Table 1

Antibody Target		Antibody Source	
Target	Abbreviation	Source	Abbreviation
Bone	-o- or -os-	Chimeric	-xi-
Cardiovascular	-c- or -ci-	Human	-u-
Immunomodulating	-l- or -li-	Humanized	-zu-
Interleukin	-k- or -ki-	Rat	-a-
Tumor	-t- or -tu-		
Virus	-v- or -vi-		

Based on the components of its name, what does trastuzumab target and what is it derived from?

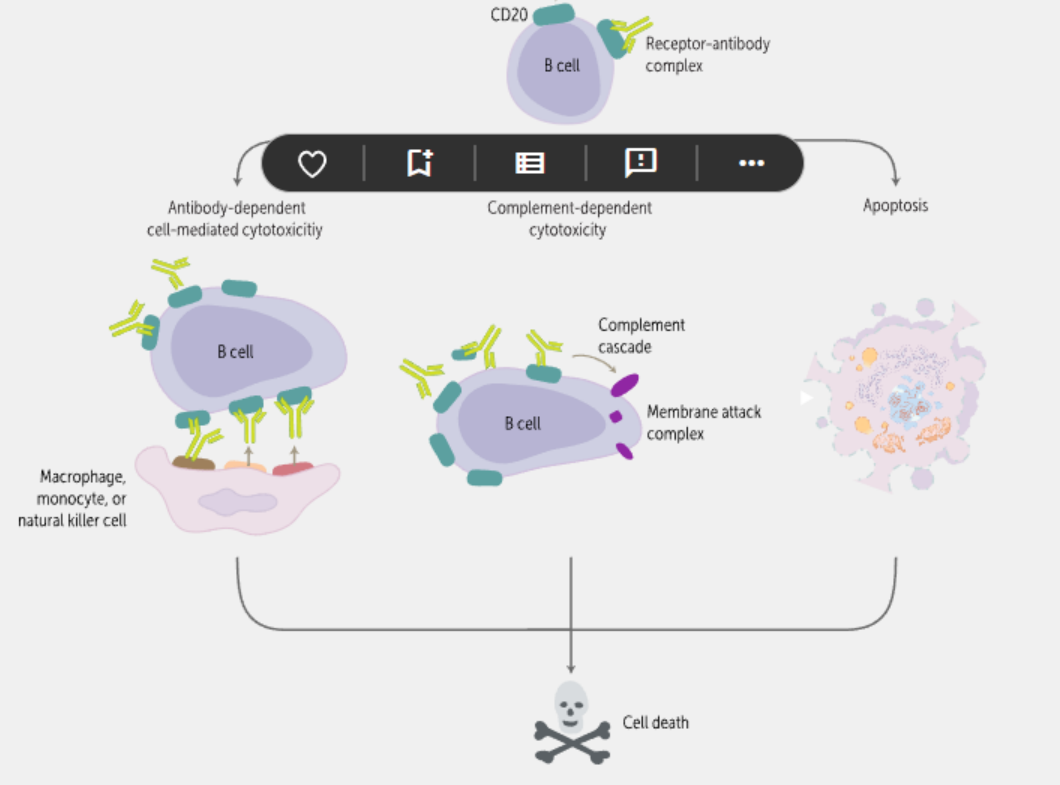
We are going to discu mmonly used in treating lymphomas and leukemia

So far, we've focused on targeted interruption of pathways within cancer cells so that they stop growing and undergo apoptosis. But there are other ways to kill cancer cells, such as by promoting an immune system attack on the cancer cells or depriving them of oxygen and blood supply.

RituxiMAB

How does Rituximab work? Rituximab is a monoclonal antibody against the **cell surface protein CD20**. CD20 is a glycosylated tetra-transmembrane phosphoprotein with intracellular N- and C-terminal regions. CD20 also has a small and a large extracellular loop that is widely expressed on developing B-cell and many B-cell malignancies. The protein normally functions to promote proliferation of B-cells, but its normal ligand is not known.

The binding of the CD20 protein by rituximab **reduces B-cell proliferation** and promotes apoptosis. But that's not all it does: it induces activation of complement, phagocytosis of the B-cell by macrophages, and cytotoxic attack by natural killer cells (Figure 3). This allows the body's immune system to attack the malignant B-cell.


Rituximab antibody
Antibody binding
CD20
B cell
Receptor-antibody complex
Antibody-dependent cell-mediated cytotoxicitiy
Complement-dependent cytotoxicity
Apoptosis
B cell
Macrophage, monocyte, or natural killer cell
Complement cascade
B cell
Membrane attack complex
Cell death

QUIZ

Figure 3

What are the adverse effects of Rituximab? Rituximab is associated with several toxicities, including potentially life-threatening infusion reactions. Fever, chills, throat itching, urticaria, and mild hypotension, they can be managed with reduced infusion rates and antihistamines. Rituximab may **reactivate hepatitis B virus** and other viral infections (e.g., cytomegalovirus, herpes simplex virus, West Nile), potentially leading to fulminant hepatitis. Patients should be screened for prior hepatitis B infection before starting therapy and monitored for signs of hepatitis B reactivation for up to 24 months after completing Rituximab treatment. Severe mucocutaneous skin reactions, including Stevens-Jo ted.

Next we are going to look at a drug that is not a TKI or a monoclonal antibody. These are small molecules that selectively target degradation of selective proteins. These agents can be classified as modulators of protein degradation. Examples of agents in this class include thalidomide, lenalidomide, and bortezomib. We will discuss bortezomib. (Thalidomide and lenalidomide are discussed in a separate brick)

How do modulators of protein degradation (Bortezomib) work? (LO#3 and 4)

BorteZOMIB is used in the treatment of multiple myeloma and mantle cell lymphoma. It inhibits the action of a crucial intracellular protein complex, the **proteasome**, which is essential for the breakdown of proteins in the cell, either to get rid of inactive or old molecules or to cut a molecule and create a new active molecule. The -zomib suffix is for drugs that inhibit proteosome degradation.

How does Bortezomib work? Bortezomib binds the **26S proteasome and inhibits** the production of **NF-κB** (nuclear factor kappa-light-chain-enhancer of activated B-cells). NF-κB is a transcription factor critical to initiation of the cell response to cell damage and promotion of cell survival. Without NF-κB, the cell is not able to grow or survive apoptosis. The molecule is initially sequestered in the cytoplasm by the IκB molecule. The proteasome degrades IκB, allowing NF-κB to be free to enter the nucleus and initiate the transcription of additional factors to keep the cell alive. Bortezomib inhibits the proteasome and prevents degradation of IκB, which means less NF-κB enters the nucleus (Figure 4).

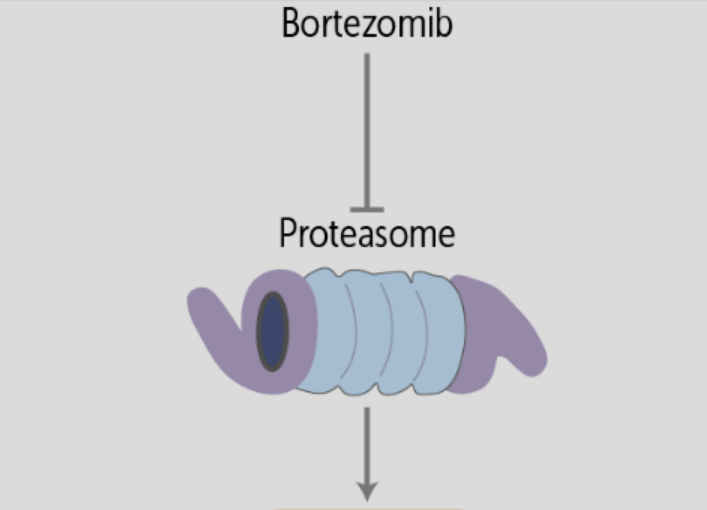


Figure 4

Bortezomib also promotes the sensitivity of tumor cells to other cytotoxic drugs, such as the anthracyclines, and prevents adhesion of multiple myeloma cells to bone marrow.

What are the potential adverse effects? Thrombocytopenia, fatigue, GI effects (nausea, vomiting, constipation, diarrhea), anemia, peripheral neuropathy, headache, and limb pain. Thrombocytopenia and neuropathy are most common and occur in 28% and 12% of patients, respectively. Dose reductions or discontinuation of Bortezomib *decreases* the neuropathic symptoms.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back on LS, how can you explain what happened to LS?

LS has been on a single dose of Rituximab for the past two years. Rituximab is a monoclonal antibody targeting CD20, which can decrease B-cell proliferation. At the same time, Rituximab can also reactivate viral infections, such as Hepatitis B (HBV) and herpes simplex virus (HSV). In some cases, such as LS, it can lead to fulminant hepatitis. In some cases, it can take 24 months for the development of Hepatitis B. Hence, routine blood work that includes liver function tests, among other laboratory tests, should be performed. Use of antivirals can be used to treat her Hepatitis infection.

Summary

- Targeted cancer therapies focus on inhibiting oncogenic pathways responsible for uncontrolled cancer cell growth and survival.
- Small molecules and monoclonal antibodies are key tools in targeting specific proteins or pathways in cancer management.
- Tyrosine kinase inhibitors (TKIs) are small molecules that block receptor function intracellularly, preventing phosphorylation and interrupting signaling pathways.
- Monoclonal antibodies attach to specific cell surface proteins, blocking signaling and potentially attracting other substances to aid in cancer cell destruction.
- Bortezomib is a p tein degradation, affecting cell survival pathways and enhancing sensitivity to other cytotoxic drugs.
- Adverse effects of these therapies can include fluid retention, QT prolongation, infusion reactions, and thrombocytopenia, among others.

Review Questions

1. Which statement describes tyrosine kinase inhibitors (TKIs) developed for targeted cancer therapy?

A. ○ TKsI have selective targets
B. ○ TKIs target cell surface receptors
C. ○ TKIs have specific effects
D. ● TKIs target intracellular molecules

Hide Explanation

The correct answer is TKIs target intracellular molecules (D). TKIs are small molecules that can cross the cell membrane to target intracellular molecules. Answer choice (A) is incorrect since TKIs lack selectivity and specificity as compared to monoclonal antibodies. Answer choice (B) is incorrect as TKIs bind to intracellular receptors. Answer choice (C) is incorrect since TKIs have multiple targets and effects.

2. Which of the following drugs targets BCR-ABL positive cancers?

A. ○ Bortezomib
B. ● Imatinib
C. ○ Cyclophosphamide
D. ○ Rituximab

Hide Explanation

The correct answer is Imatinib (B). Imatinib is a tyrosine kinase inhibitor that targets the BCR-ABL protein. Bortezomib (A) is a modulator of protein degradation. Cyclophosphamide (C) is a alkylating agent and does not target the BCR-ABL gene. Rituximab (D) is a CD20 inhibitor.

3. Which of the following could likely reactivate Hepatitis B infection?

A. ○ Dasatinib
B. ● Rituximab
C. ○ Imatinib
D. ○ Bortezomib

Hide Explanation

The correct answer is Rituximab (D). Rituximab is a CD20 inhibitor that targets B-cell proliferation. Rituximab can reactivate Hepatitis B infection up to 24 months of discontinuation. Dasatinib (A) and Imatinib (C) are BCR-ABL inhibitors that are more likely to alter liver function tests and cause fluid retention. Bortezomib (D) is a modulator of protein degradation that causes thrombocytopenia as an adverse effect.

4. Which agent targets the degradation of the proteosomes?

A. ○ Rituximab
B. ○ Imatinib
C. ● Bortezomib
D. ○ Dasatinib

Hide Explanation

The correct answer is Bortezomib (C). Bortezomib targets proteosome degradation and breakdown of IκB, preventing the production of NF-κB. This decreases cell proliferation. Answer choice (A) Rituximab is a monoclonal antibody that targets CD20 to inhibit cell proliferation. Choice (B) Imatinib and choice (D) Dasatinib are TKIs that inhibit BCR ll proliferation.

proliferation. Answer choice (A) Rituximab is a monoclonal antibody that targets CD20 to inhibit cell proliferation. Choice (B) Imatinib and choice (D) Dasatinib are TKIs that inhibit BCR-ABL function and decrease downstream cell proliferation.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Compare and contrast the key features between Tyrosine kinase inhibitors and monoclonal antibodies~~
02 ☑ ~~Explain the mechanism of action of targeted inhibitors used in cancer therapy.~~
03 ☑ ~~Explain the mechanism of action of modulators of protein degradation~~
04 ☑ ~~Describe the adverse effects associated with the use of targeted inhibitors and strategies for mitigating these effects.~~

Objective Confidence: 4/4